leveraged-etfs-returns-dispersion

January 27, 2026

1 How do the long-term returns of leveraged ETFs diverge from their underlying index?

1.1 Python Imports

[1]:
```
# Standard Library
import datetime
import io
import os
import random
import sys
import warnings

from datetime import datetime, timedelta
from pathlib import Path

# Data Handling
import numpy as np
import pandas as pd

# Data Visualization
import matplotlib.dates as mdates
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
from matplotlib.ticker import FormatStrFormatter, FuncFormatter, MultipleLocator

# Data Sources
import yfinance as yf
import pandas_datareader.data as web

# Statistical Analysis
import statsmodels.api as sm

# Machine Learning
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
```

```
# Suppress warnings
warnings.filterwarnings("ignore")
```

1.2 Add Directories To Path

[2]:
```
# Add the source subdirectory to the system path to allow import config from␣
↪settings.py
current_directory = Path(os.getcwd())
website_base_directory = current_directory.parent.parent.parent
src_directory = website_base_directory / "src"
sys.path.append(str(src_directory)) if str(src_directory) not in sys.path else␣
↪None

# Import settings.py
from settings import config

# Add configured directories from config to path
SOURCE_DIR = config("SOURCE_DIR")
sys.path.append(str(Path(SOURCE_DIR))) if str(Path(SOURCE_DIR)) not in sys.path␣
↪else None

# Add other configured directories
BASE_DIR = config("BASE_DIR")
CONTENT_DIR = config("CONTENT_DIR")
POSTS_DIR = config("POSTS_DIR")
PAGES_DIR = config("PAGES_DIR")
PUBLIC_DIR = config("PUBLIC_DIR")
SOURCE_DIR = config("SOURCE_DIR")
DATA_DIR = config("DATA_DIR")
DATA_MANUAL_DIR = config("DATA_MANUAL_DIR")

# Print system path
for i, path in enumerate(sys.path):
    print(f"{i}: {path}")
```

```
0: /usr/lib/python313.zip
1: /usr/lib/python3.13
2: /usr/lib/python3.13/lib-dynload
3:
4: /home/jared/python-virtual-envs/general-venv-p313/lib/python3.13/site-
packages
5: /home/jared/python-virtual-envs/general-venv-p313/lib/python3.13/site-
packages/setuptools/_vendor
6:
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io_congo/src
```

1.3 Track Index Dependencies

[3]:
```
# Create file to track markdown dependencies
dep_file = Path("index_dep.txt")
dep_file.write_text("")
```

[3]: 0

1.4 Python Functions

[4]:
```
from build_index import build_index
from df_info import df_info
from df_info_markdown import df_info_markdown
from export_track_md_deps import export_track_md_deps
from load_data import load_data
from pandas_set_decimal_places import pandas_set_decimal_places
from plot_timeseries import plot_timeseries
from summary_stats import summary_stats
from yf_pull_data import yf_pull_data
```

[5]:
```
# Set decimal places
pandas_set_decimal_places(2)
```

1.5 Data Overview

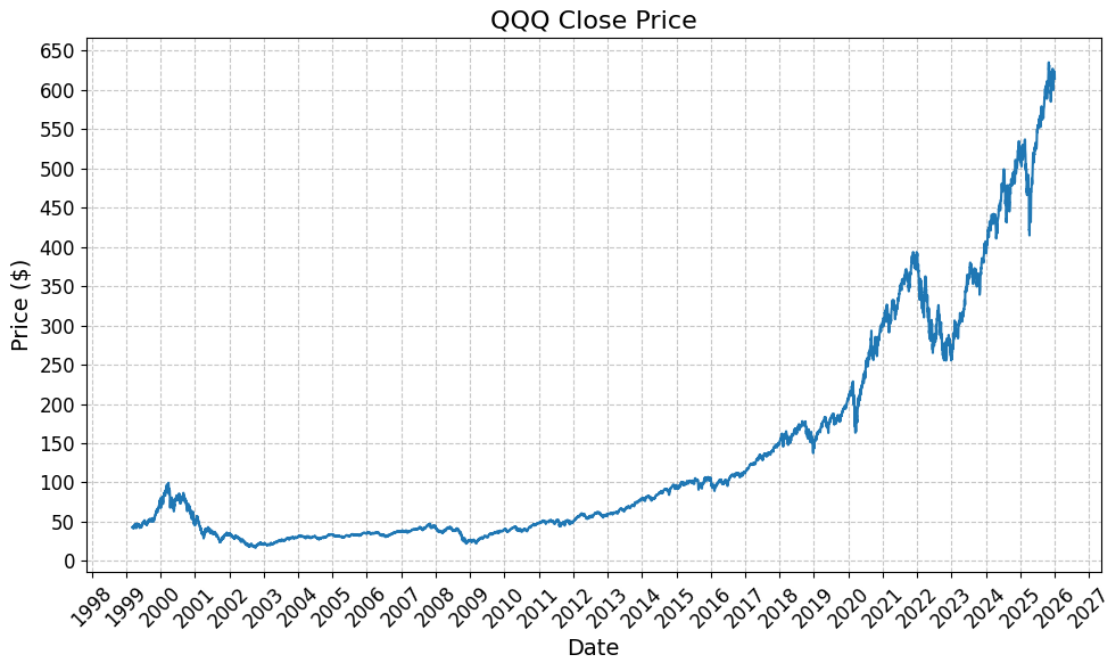
1.5.1 Acquire & Plot Data (QQQ)

[6]:
```
yf_pull_data(
    base_directory=DATA_DIR,
    ticker="QQQ",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    excel_export=True,
    pickle_export=True,
    output_confirmation=True,
)

qqq = load_data(
    base_directory=DATA_DIR,
    ticker="QQQ",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    timeframe="Daily",
    file_format="pickle",
)

# Rename columns to "QQQ_Close", etc.
qqq = qqq.rename(columns={
```

```
    "Close": "QQQ_Close",
    "High": "QQQ_High",
    "Low": "QQQ_Low",
    "Open": "QQQ_Open",
    "Volume": "QQQ_Volume"
})
```

```
[*********************100%***********************]  1 of 1 completed


The first and last date of data for QQQ is:

            Close  High   Low  Open   Volume
Date
1999-03-10  43.13 43.21 42.47 43.18  5232000

            Close   High    Low   Open    Volume
Date
2026-01-26 625.46 627.61 622.12 623.21  35983000

Yahoo Finance data complete for QQQ
--------------------
```

[7]:
```
plot_timeseries(
    price_df=qqq,
    plot_start_date="1999-03-10",
    plot_end_date="2025-12-31",
    plot_columns=["QQQ_Close"],
    title="QQQ Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=50,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="01_QQQ_Price",
)
```



[8]:
```
# Copy this <!-- INSERT_01_QQQ_Price_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="01_QQQ_Price.md",
    content=df_info_markdown(df=qqq, decimal_places=2),
    output_type="markdown",
)
```

```
Exported and tracked: 01_QQQ_Price.md
```

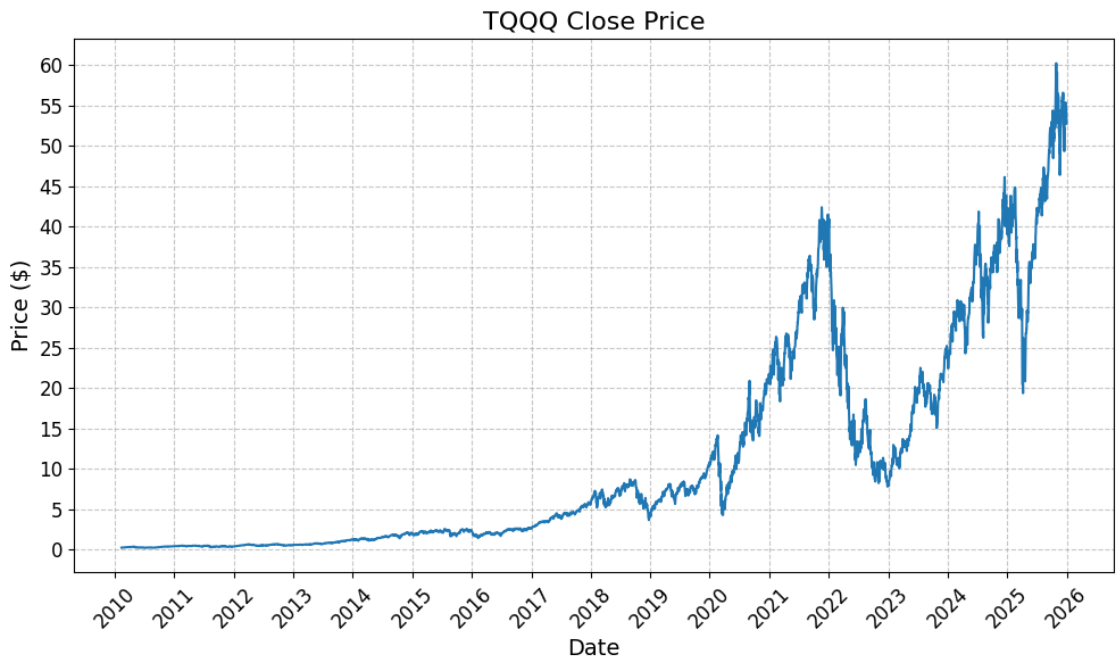
1.5.2 Acquire & Plot Data (TQQQ)

[9]:
```
yf_pull_data(
    base_directory=DATA_DIR,
    ticker="TQQQ",
    source="Yahoo_Finance",
    asset_class="Exchange_Traded_Funds",
    excel_export=True,
    pickle_export=True,
    output_confirmation=True,
)

tqqq = load_data(
    base_directory=DATA_DIR,
    ticker="TQQQ",
    source="Yahoo_Finance",
```

```
    asset_class="Exchange_Traded_Funds",
    timeframe="Daily",
    file_format="pickle",
)

# Rename columns to "TQQQ_Close", etc.
tqqq = tqqq.rename(columns={
    "Close": "TQQQ_Close",
    "High": "TQQQ_High",
    "Low": "TQQQ_Low",
    "Open": "TQQQ_Open",
    "Volume": "TQQQ_Volume"
})
```

```
[*********************100%***********************]  1 of 1 completed


The first and last date of data for TQQQ is:

            Close  High  Low  Open   Volume
Date
2010-02-11   0.21  0.21 0.19  0.19  6912000

            Close  High   Low  Open    Volume
Date
2026-01-26  55.08 55.63 54.21 54.47  58084300

Yahoo Finance data complete for TQQQ
--------------------
```

[10]:
```
plot_timeseries(
    price_df=tqqq,
    plot_start_date="2010-02-11",
    plot_end_date="2025-12-31",
    plot_columns=["TQQQ_Close"],
    title="TQQQ Close Price",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Price ($)",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=5,
    grid=True,
    legend=False,
    export_plot=True,
    plot_file_name="02_TQQQ_Price",
)
```



[11]:
```
# Copy this <!-- INSERT_02_TQQQ_Price_HERE --> to index_temp.md
export_track_md_deps(
    dep_file=dep_file,
    md_filename="02_TQQQ_Price.md",
    content=df_info_markdown(df=qqq, decimal_places=2),
    output_type="markdown",
)
```

```
Exported and tracked: 02_TQQQ_Price.md
```

1.5.3 Calculate Cumulative Returns & Drawdowns (QQQ & TQQQ)

[12]:
```
etfs = ["QQQ", "TQQQ"]

# Merge dataframes and drop rows with missing values
qqq_tqqq_aligned = tqqq.merge(qqq, left_index=True, right_index=True,␣
↪how='left')
qqq_tqqq_aligned = qqq_tqqq_aligned.dropna()

# Calculate cumulative returns
for etf in etfs:
    qqq_tqqq_aligned[f"{etf}_Return"] = qqq_tqqq_aligned[f"{etf}_Close"].
↪pct_change()
    qqq_tqqq_aligned[f"{etf}_Cumulative_Return"] = (1 +␣
↪qqq_tqqq_aligned[f"{etf}_Return"]).cumprod() - 1
```

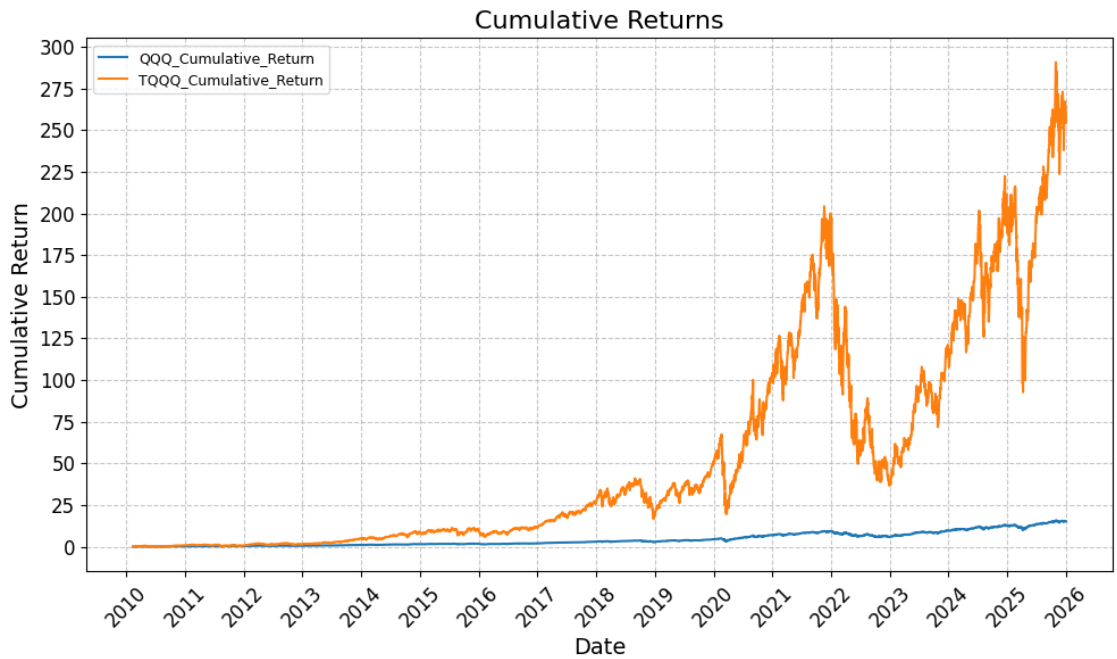
```
    qqq_tqqq_aligned[f"{etf}_Cumulative_Return_Plus_One"] = 1 +␣
↪qqq_tqqq_aligned[f"{etf}_Cumulative_Return"]
    qqq_tqqq_aligned[f"{etf}_Rolling_Max"] =␣
↪qqq_tqqq_aligned[f"{etf}_Cumulative_Return_Plus_One"].cummax()
    qqq_tqqq_aligned[f"{etf}_Drawdown"] =␣
↪qqq_tqqq_aligned[f"{etf}_Cumulative_Return_Plus_One"] /␣
↪qqq_tqqq_aligned[f"{etf}_Rolling_Max"] - 1
    qqq_tqqq_aligned.drop(columns=[f"{etf}_Cumulative_Return_Plus_One",␣
↪f"{etf}_Rolling_Max"], inplace=True)
```

[13]:
```
qqq_tqqq_aligned
```

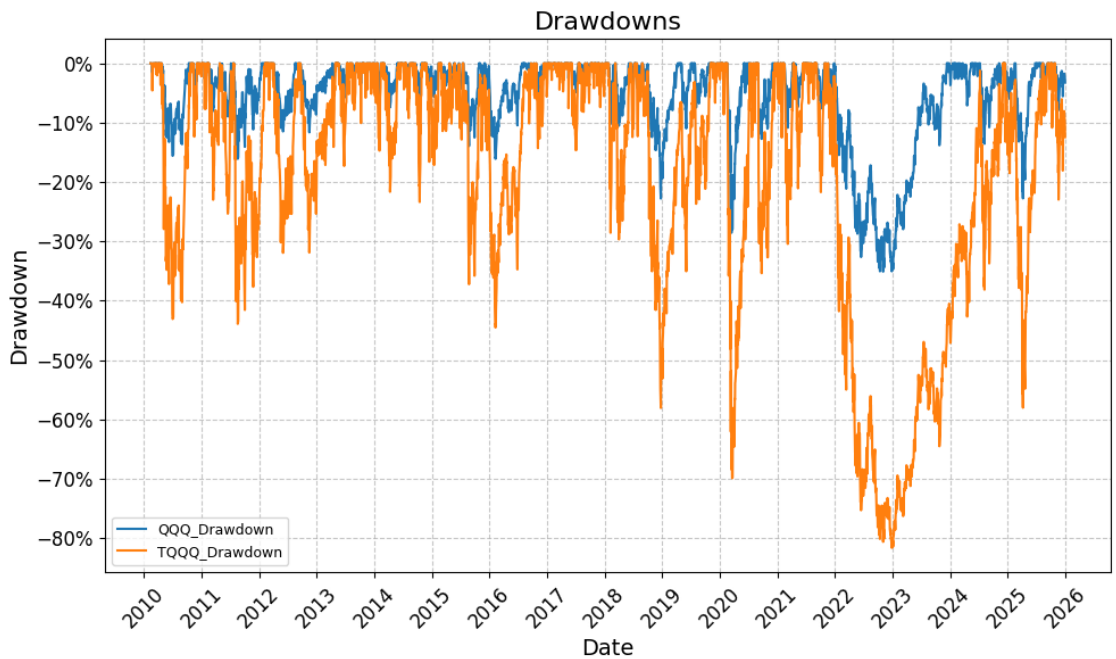
[13]:
```
            TQQQ_Close  TQQQ_High  TQQQ_Low  TQQQ_Open  TQQQ_Volume  \
Date
2010-02-11        0.21       0.21      0.19       0.19      6912000
2010-02-12        0.21       0.21      0.20       0.20     17203200
2010-02-16        0.22       0.22      0.21       0.21     19238400
2010-02-17        0.22       0.22      0.21       0.22     38361600
2010-02-18        0.22       0.22      0.22       0.22     77721600
…                  …          …         …          …           …
2026-01-20       50.73      52.56     50.48      51.39    129209100
2026-01-21       52.73      53.80     50.67      51.07    150230100
2026-01-22       53.90      54.33     53.15      54.32     90992300
2026-01-23       54.38      55.07     53.32      53.59     78066100
2026-01-26       55.08      55.63     54.21      54.47     58084300

            QQQ_Close  QQQ_High  QQQ_Low  QQQ_Open  QQQ_Volume  QQQ_Return  \
Date
2010-02-11      37.95     38.06    37.16     37.40    93050800         NaN
2010-02-12      38.03     38.13    37.51     37.61    93938500        0.00
2010-02-16      38.52     38.54    38.11     38.31    83861400        0.01
2010-02-17      38.73     38.73    38.46     38.69    82691700        0.01
2010-02-18      38.98     39.05    38.63     38.71    65752600        0.01
…                 …         …        …         …           …           …
2026-01-20     608.06    615.06   607.05    610.53    81988900       -0.02
2026-01-21     616.28    620.42   607.86    609.47    79837900        0.01
2026-01-22     620.76    622.46   617.78    622.35    42254800        0.01
2026-01-23     622.72    625.40   618.65    619.73    43645800        0.00
2026-01-26     625.46    627.61   622.12    623.21    35983000        0.00

            QQQ_Cumulative_Return  QQQ_Drawdown  TQQQ_Return  \
Date
2010-02-11                    NaN           NaN          NaN
2010-02-12                   0.00          0.00         0.00
2010-02-16                   0.01          0.00         0.04
2010-02-17                   0.02          0.00         0.02
2010-02-18                   0.03          0.00         0.02
```

```
…                                …            …            …
2026-01-20                    15.02          -0.04        -0.06
2026-01-21                    15.24          -0.03         0.04
2026-01-22                    15.36          -0.02         0.02
2026-01-23                    15.41          -0.02         0.01
2026-01-26                    15.48          -0.01         0.01

            TQQQ_Cumulative_Return  TQQQ_Drawdown
Date
2010-02-11                     NaN            NaN
2010-02-12                    0.00           0.00
2010-02-16                    0.04           0.00
2010-02-17                    0.06           0.00
2010-02-18                    0.08           0.00
…                              …             …
2026-01-20                  244.79          -0.16
2026-01-21                  254.48          -0.12
2026-01-22                  260.15          -0.10
2026-01-23                  262.47          -0.10
2026-01-26                  265.87          -0.09

[4013 rows x 16 columns]
```

[14]:
```
plot_timeseries(
    price_df=qqq_tqqq_aligned,
    plot_start_date="2010-02-11",
    plot_end_date="2025-12-31",
    plot_columns=["QQQ_Cumulative_Return", "TQQQ_Cumulative_Return"],
    title="Cumulative Returns",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Cumulative Return",
    y_format="Decimal",
    y_format_decimal_places=0,
    y_tick_spacing=25,
    grid=True,
    legend=True,
    export_plot=True,
    plot_file_name="03_Cumulative_Returns",
)
```



[15]:
```
plot_timeseries(
    price_df=qqq_tqqq_aligned,
    plot_start_date="2010-02-11",
    plot_end_date="2025-12-31",
    plot_columns=["QQQ_Drawdown", "TQQQ_Drawdown"],
    title="Drawdowns",
    x_label="Date",
    x_format="Year",
    x_tick_rotation=45,
    y_label="Drawdown",
    y_format="Percentage",
    y_format_decimal_places=0,
    y_tick_spacing=0.10,
    grid=True,
    legend=True,
    export_plot=True,
    plot_file_name="03_Drawdowns",
)
```



[16]:
```
qqq_sum_stats = summary_stats(
    fund_list=["QQQ"],
    df=qqq_tqqq_aligned[["QQQ_Return"]],
    period="Daily",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)

tqqq_sum_stats = summary_stats(
    fund_list=["TQQQ"],
    df=qqq_tqqq_aligned[["TQQQ_Return"]],
    period="Daily",
    use_calendar_days=False,
    excel_export=False,
    pickle_export=False,
    output_confirmation=False,
)

# tlt_sum_stats = summary_stats(
#     fund_list=["Portfolio", "SPY", "TLT"],
#     df=portfolio_monthly[["TLT_Monthly_Return"]],
#     period="Monthly",
#     use_calendar_days=False,
```

```
#     excel_export=False,
#     pickle_export=False,
#     output_confirmation=False,
# )

sum_stats = qqq_sum_stats.combine_first(tqqq_sum_stats)
```

[17]:
```
sum_stats
```

[17]:
```
              Annualized Mean  Annualized Volatility  Annualized Sharpe Ratio  \
QQQ_Return               0.20                   0.21                     0.96
TQQQ_Return              0.54                   0.61                     0.88

              CAGR  Daily Max Return Daily Max Return (Date)  Daily Min Return  \
QQQ_Return    0.19              0.12              2025-04-09             -0.12
TQQQ_Return   0.42              0.35              2025-04-09             -0.34

             Daily Min Return (Date)  Max Drawdown       Peak     Trough  \
QQQ_Return               2020-03-16         -0.35 2021-12-27 2022-11-03
TQQQ_Return              2020-03-16         -0.82 2021-11-19 2022-12-28

             Recovery Date  Days to Recover  MAR Ratio
QQQ_Return     2023-12-13              405       0.55
TQQQ_Text      2024-12-04              707       0.51
```

[18]:
```
# # Copy this <!-- INSERT_05_Portfolio_Stats_DF_HERE --> to index_temp.md
# export_track_md_deps(dep_file=dep_file, md_filename="05_Portfolio_Stats_DF.
↪md", content=sum_stats.to_markdown(floatfmt=".3f"))
```